Arista Access Node MAR-ACC-49 Hardware/Environmental Fault Recovery Procedure

Node: MAR-ACC-49 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1990

Description

This Standard Operating Procedure (SOP) outlines the steps to recover from a hardware/environmental fault on the Arista Access Node MAR-ACC-49 at the Marathahalli Core site. The fault is characterized by ASIC memory parity errors, line card crashes, or power supply redundancy loss, triggering the ALARM_1990 alarm with a minor severity. The root cause of this failure domain resides in the hardware components of the access node, particularly the ASICs, line cards, and power supplies, which are critical to the node's data forwarding capabilities. The Arista EOS software, running on the node, relies on these hardware components to process and forward traffic. When a hardware fault occurs, it can lead to data corruption, packet drops, or even a complete loss of connectivity. The EOS software will automatically detect the fault and trigger the ALARM_1990 alarm, prompting the network administrator to investigate and resolve the issue. This SOP provides a structured approach to diagnosing and resolving the fault, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Marathahalli Core site to physically access the access node.

- Verify the redundant link status and ensure it is functioning properly.

- Check the power supply redundancy and ensure it is operational.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal error logs, Expected Output: Error logs indicating ASIC memory parity errors, line card crashes, or power supply redundancy loss., Description: This command displays the system internal error logs, providing insight into the fault's nature and severity.

Command: show hardware, Expected Output: Hardware information, including ASICs, line cards, and power supplies., Description: This command displays the hardware information, allowing the administrator to verify the presence and functionality of the affected components.
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Command: show redundancy, Expected Output: Redundancy information, including power supply redundancy status., Description: This command displays the redundancy information, enabling the administrator to verify the power supply redundancy status.
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Resolution Steps

Step 1: Step: Power Cycle the Access Node, Command: reload, Description: Perform a power cycle of the access node to reset the hardware components and resolve any transient faults.

Step 2: Step: Replace Faulty ASICs or Line Cards, Command: no switch , Description: Replace the faulty ASICs or line cards with new ones, ensuring optimal data forwarding performance.

Step 3: Step: Verify Power Supply Redundancy, Command: show redundancy, Description: Verify the power supply redundancy status, ensuring it is operational and functioning properly.

Post-Check Verification

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Command: show system internal error logs, Expected Output: No error logs indicating ASIC memory parity errors, line card crashes, or power supply redundancy loss., Description: Verify that the error logs no longer indicate the fault, ensuring the issue has been resolved.

Command: show hardware, Expected Output: Hardware information indicating the replacement or repair of faulty components., Description: Verify that the hardware information indicates the replacement or repair of the faulty components.

Command: show redundancy, Expected Output: Redundancy information indicating a fully operational power supply redundancy., Description: Verify that the redundancy information indicates a fully operational power supply redundancy.
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Roll-back Procedures

- Step: Restore Original Configuration, Command: configure replace , Description: Restore the original configuration to ensure the node functions as intended.

- Step: Revert to Previous Firmware Version, Command: firmware rollback, Description: Revert to the previous firmware version to ensure compatibility with the affected hardware components.